

DIPAN BARTAULA

dipan.bartaula@naamii.org.np | deepanzcreed001@gmail.com | +977-9742559750
Pulchowk, Lalitpur, Nepal | linkedin.com/in/dipan-bartaula | github.com/DipanBartaula

PROFESSIONAL SUMMARY

AI Engineer with about 3 years of industry, research, and freelance experience across LLMs and Computer Vision. Builds production LLM systems (multi-agent workflows, MCP and tool calling, RAG, evals, guardrails) as Python services, and vision and generative models across 3D vision, virtual try-on, and diffusion. Trains models with verifiable-reward RL and ships them from distributed training to cloud and edge. First author of two papers under review, including CURVTON-205K, the largest virtual try-on benchmark.

EDUCATION

BE Computer Engineering | IOE, Pulchowk Campus — TU, Nepal 2022 – 2026

Coursework: Engineering Mathematics, Artificial Intelligence, Theoretical Computer Science, and Applied Computer Science.

Training: Big Data Engineering, Generative AI.

High School (+2 Science) | Hetauda School of Management and Social Sciences 2019 – 2021

Core subjects: Physics, Mathematics, Chemistry, and Computer Science.

ACHIEVEMENTS

- IEEEExtreme 2024 Winner — **Rank 1 in Nepal** out of all participating university teams
- Patternverse Competition (Alternative Technology) — **Top 5 Finalist** among 50+ competing teams
- Semicolon MLH Hackathon — **Top 10 Finalist** among 70+ competing teams
- Samsung Innovation Campus — Admitted among **Top 30 students nationally**
- **Reviewer** — Peer-reviewed AI/ML Conference, IOE Pulchowk Campus
- NAAMII Ultrasound AI Hackathon — Selected for competitive international medical AI challenge

RESEARCH PAPERS (First Author · Cumulative Compute: 5,000 H200 GPU Hours, ≈ \$23K)

CURVTON-205K: A Scalable Mask-Free Synthetic Dataset for Virtual Try-On

Largest mask-free virtual try-on benchmark (205K+ samples) with difficulty-tiered evaluation; released on Hugging Face Hub.

DreamCloth: Inverse Estimation of Cloth Simulation Parameters via Video Priors

Recovers cloth mass, stiffness, and friction from monocular video via video diffusion priors and differentiable physics.

Foresight: Anticipatory Computation for Streaming Vision-Language Models

Training-free dual-stream Siamese LLMs that let streaming VLMs anticipate upcoming context and adapt compute online.

EXPERIENCE

Research Engineer | NAAMII / Fogsphere Multimodal Learning Lab Jan 2026 – Present (10 months)

Research Edge AI, world models, and medical imaging, stress-testing SOTA multimodal models; run distributed SLURM/AWS training (mixed precision, gradient checkpointing, MLflow/W&B) with cost-aware GPUs on RunPod, Vast.ai, and Lightning AI.

Founding ML Engineer (Part-time) | AI Lab Pvt. Ltd. Dec 2025 – Present (11 months)

Building streaming, on-device Nepali-English-Medical code-mixed ASR: designed the low-latency inference pipeline, distilled edge ASR backbones without accuracy loss, and built large-scale SFT data pipelines and a custom benchmark.

Junior AI Engineer | Accelerated Komputing Jun 2025 – Dec 2025 (7 months)

Architected DAG-based multi-agent LLM orchestration that breaks engineering queries into reasoning graphs for CAD/CAE automation. Built closed-loop LLM-FEM pipelines with autonomous tool invocation and self-correction. Led NVIDIA Isaac GROOT integration with constrained decoding for human-free execution.

Freelance AI Engineer & Technical Instructor | Independent Feb 2024 – Dec 2025 (1 yr 6 months, intermittent)

Contributed to part-level 3D asset generation for an Israel-based stealth 3D-printing startup; delivered a voice-to-video system (LLM scripting, diffusion lip sync), a skin analysis pipeline (21+ conditions, severity, ingredient recommendations), and DAG-orchestrated tool-calling agents for clients. Supported paper writing and peer review; taught 255+ hours of Data Science, Python, and Agentic AI/RAG.

PROJECTS

AGENTIC AI

Agentic ML System — *Autonomous End-to-End Machine Learning Pipeline*

LangChain multi-agent system that turns a raw dataset into a deployed model from one natural-language prompt, with typed-state agents for EDA, preprocessing, feature engineering, model selection, and monitoring. Adds meta-learned algorithm selection, Bayesian (GP) tuning, KS/PSI drift-triggered retraining, and SHAP/PDP explainability, cutting manual ML effort by 70%+. LangSmith-monitored, with prompt caching, versioned system prompts and skill files, agentic evals, regression tests, and per-version token-cost tracking.

CAD/CAE Agents and RLVR — *Agentic Engineering Design & RL Gym*

- **Agentic Engineering Design:** Turns text into structural, thermal, and fluid simulations, routing each task by design complexity across models with a spectrum of coding and engineering capability; code is ontology-validated (units, boundary conditions, mesh-solver fit) before running, solver logs drive self-repair, and guardrails wrap every step. RAG over drawings, design docs, and the client's scripting framework: structure-aware chunking, BM25 + dense retrieval, reranking, recall@k/MRR evals.
- **RL Gym:** Headless, always-on Python gym where 3D foundation reward models and rule-based checks (code quality, compile and runtime success, latency, complexity, constraints) score agent-written scripts to train Qwen via QLoRA SFT and GRPO-based RLVR, with async multi-GPU rollouts and KL regularization against reward hacking. Langfuse traces evals, regressions, and token cost.

LLM-Driven Lecture Video Adaptation — *Difficulty-Aware Playback*

Django service that transcribes lectures (WhisperX), rates topic difficulty with o4-mini, and re-renders video slower on hard sections and faster on easy ones with intelligible audio; Cloudinary storage.

AI Agents Integration & Engineering — *Custom MCP, Agent Harnesses & Self-Evolving Skills*

Built custom MCP servers and an integration layer giving agents tool, SSH, and API-gateway access via structured tool calling, typed state, DAG orchestration, A2A, and constrained decoding. Harness loops checkpoint persistent state to resume long runs, route tool failures back through the reasoning graph, and distill each run's takeaways into optimized skill files. Red-teamed with YAML attack scenarios (indirect prompt injection, data exfiltration) and AI-generated attacks, co-evolving guardrails.

DIFFUSION MODELS

Video Generation Optimization — *LoRA, Agentic Refinement & RL Preference Supervision*

Optimized pretrained video generation models for lower latency and memory with temporally consistent output, and LoRA fine-tuned them on custom domain-specific data for subject consistency and visual quality. Added an agentic loop that diagnoses failed outputs and regenerates with revised settings, plus RL preference tuning with VLM rewards for alignment, temporal consistency, and quality.

Few-Step Diffusion Distillation — *LADD*

Distilled a diffusion model with Latent Adversarial Diffusion Distillation (LADD), cutting inference from 20 to 2 denoising steps.

DeepFit — *Diffusion VTON*

Built a Stable Diffusion 1.5 virtual try-on system with pose-conditioned attention, LoRA (90% fewer trainable parameters, 24GB VRAM), and appearance-flow/TPS warping, rivaling 2× compute systems; served via a Flask REST API with versioned checkpoints.

3D Asset Editing Pipeline — *Text-Driven 3D Modification*

Built a text-driven 3D editing pipeline that edits in image space (Qwen-Edit 2509, HunyuanDiT) and propagates edits to NeRF/3D Gaussian Splatting via score distillation with multi-view consensus and CLIP+SAM segmentation.

EDGE AI

On-Device Edge Stack — *Efficient RAG, Model Portability & Mobile Vision*

Built a fully on-device stack: TurboVec and TurboQuant for cost-effective RAG and inference, proactive video processing that prunes redundant compute, ONNX/TensorRT/MLX export for iOS and Android, and MobileOne (featured at CVPR 2026) fine-tuned for deepfake detection and medical imaging, with profiling-guided graph optimization for real-time budgets. Cut serving cost with FlashAttention, continuous batching, and shared prefill for same-context queries, plus edge deployment to avoid API token costs.

SKILLS

AI / ML & Computer Vision: Deep Learning, RL, Diffusion Models, LLMs, Federated Learning; pose estimation, 3D reconstruction, NeRF, 3D Gaussian Splatting, SAM, CLIP, YOLO; PyTorch, TensorFlow, scikit-learn, OpenCV

Agentic AI & RAG: Tool calling, MCP, A2A, structured outputs (Pydantic, constrained decoding), LangChain/LangGraph, LangSmith, Langfuse, evals and regression testing, prompt caching and versioning, guardrails; pgvector, Qdrant, TurboVec, hybrid retrieval, reranking

Model Deployment & Optimization: FastAPI, Django, Flask, Hugging Face; ONNX Runtime, TensorRT, Core ML/MLX, ExecuTorch, llama.cpp, OpenVINO; quantization (PTQ, QAT, AWQ, GPTQ, TurboQuant)

MLOps, Cloud & Data: AWS, Azure, Docker, Kubernetes, SLURM, DDP/FSDP, MLflow, W&B, Git; Spark, Kafka, Airflow, Databricks; Python, C++, CUDA, Bash

Languages: English (Professional) — Nepali (Native) — Hindi (Fluent)